SCORE: A Framework for Synthesizing Complete, Owned, Responsive Environments

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Abstract

Games, films and robot training need 3D environments that are complete, editable and owned by their creators; world models and one-shot generators give frames no engine opens, or one fused scene without separate objects or clear rights. Studios already build such environments by a standard: scenes layered in OpenUSD, each scene sitting within its parent scene and each department's work on it a layer of its own that can be checked and changed apart from the rest. SCORE builds environments by this convention. The creator decides from the top down: he locks an environment concept (EC), its style and its physics, which holds the scenes and their layering, and then a concept for each scene, in order of parent scene and child scene. The framework makes every layer below with open vision foundation models, procedural tools and coding agents, each layer checked before the next is made. Every model allows commercial use of its output, so the creator stays the author. We describe the framework with its first environment as the worked example; its results are marked gaps until its remake has run. Code and paper are available at https://github.com/Babon-Innovations-b-v/score (MIT license).

Gap: Figure 1, one environment top down. The EC of environment 1 (its style and its physics), a top scene on its site, a parent scene and its child scenes, and one scene's layers side by side. It waits on the remake's batch.

1 Introduction

Interactive 3D environments are the raw material of games, film and embodied AI [1], and building one is expensive: every object is modelled, surfaced, placed, given collision, lit and given sound in one style, and the environment stays editable. Generative models promise to take over much of this, but in GDC's 2026 survey 52 percent of game workers thought generative AI was hurting the industry [2]; a tool for building environments has to leave the result in the creator's hands.

Studios already have a standard for this: OpenUSD [3], the scene description Pixar made for its films. Each artist works in a file of their own, a layer; the layers are stacked in a stated order, so departments work on one scene without editing each other's work, and scenes are assembled from smaller ones by reference. In this convention a scene sits within its parent scene and holds its child scenes, and each department's work on it (layout, set dressing, surfaces, light, effects, motion, sound) is a layer that can be checked and changed apart from the rest. SCORE builds environments by this convention, with the creator deciding and the framework doing the departments' work. The creator locks an environment concept (EC): its style and its physics. The EC holds the scenes and their layering. For each scene a concept is then locked, in order of parent scene and child scene, and from it the framework makes every layer below, each its own OpenUSD layer, checked before the next is made.

World models generate an environment as frames [4], [5] that no engine can open. 3D environment generators [6], [7] produce geometry, but mostly as one fused asset, from which separate objects are recovered after the fact [8], [9]. Both work in one shot, so a creator cannot find or change the step behind a fault. Procedural systems [10], [11] and coding agents that write scenes as programs [12], [13], [14], [15], [16], [17] keep the environment explicit, but rebuild a scene that already exists or draw their objects from a fixed library; their models often float or come apart [18], and such agents judge their own geometry unreliably [19]; on scenes rebuilt from video, the best model scores 0.91 on static scenes but 0.67 on dynamic ones [20], which is why SCORE checks motion as a layer of its own. Open vision foundation models [21] now read shapes [22], masks [23], depth [24] and parts [25] out of pictures, and a coding agent can connect them as tools; none of them provides one style across objects that different models made, which the field names as an open problem [26], [27].

SCORE's environments are meant to be **complete**: every object, surface, light and sound exists wherever the viewer looks. They are **owned**: every model in the framework allows commercial use of its output, so the creator stays the author, holding the full commercial rights. They are meant to be **responsive**: every object has real collision, mass and material. And they can be any kind of **environment**, from a house on a plot of ground to a system of worlds. To our knowledge, no other openly available environment generation work joins vision foundation models into engine-ready environments that are built from reference pictures, steered by a person's choices at each step, and made only with models that allow commercial use of their output.

We contribute **the SCORE framework**, which builds an environment by the studios' convention, the creator deciding top down and the framework making every layer below (Section 2); **measured gates** for every layer of a scene and a completion gate over them (Section 3); and **evidence from a first environment**, left as gaps until its remake has run (Section 4).

2 The framework

The creator works in the creator cockpit, one signed-in web page: there he decides the design, buys paid work and reviews what the framework reports. Below him, an environment is a table of stages; environment 1's has 149 rows. A **node** is one stage for one scene (`assembly@hangar`) or for the environment, and one runner makes, judges and accepts every node. From an approved design everything runs without a further yes.

2.1 Keys, gates and accepts

Keys. A node is made only when its key has no finish record. The key hashes the stage and scene, a code version, the contracts of its output and inputs, its parameters, a seed derived from the environment's, the pinned model images, every environment file it reads and the accepted output of every node it reads. A fault fixed in one stage moves only its key and those after it. Outputs are kept by content hash, never overwritten.

Gates and accepts. Each gate answers pass, fail or unknown, with measures and a reason; missing evidence is unknown, and unknown blocks like fail. The runner accepts an output by itself when every gate passed and it is no worse than the one accepted before: no answer dropped, no measure moved the wrong way by a step a viewer could see, no gate missing. Later nodes read only accepted outputs. The creator keeps four actions, buttons in the cockpit: reject an accepted output (it moves back to the one before, and every node that read it is stale), accept a worse one with a reason, waive one gate for one key, and override the runner for one paid run. A barrier keeps agents from them:

a store that refuses the creator's records from an agent's process, and a hook.

Batches. Paid work is bought as a batch, with one estimate and one ETA; the creator's yes to that estimate is the only spend decision, and the batch stops at its high estimate plus 10 percent. Inside a batch every paid job runs once, with no retry; a failed output becomes a batch gap that the completion gate lists, and spending more on it is a new batch. Each node's review page is built from the store alone, and the creator reviews a scene in a 3D viewer on the store's copy of its accepted assembly.

2.2 From concept to scene

The environment concept. The creator first locks the EC, in a short interview with Gemini 3.1 Pro [28] in the cockpit: its style and its **physics**, with his reference pictures. The EC holds the scenes and their layering: the top scenes and the child scenes each may hold. Its physics is what everything stands on and in and how it moves, at the scale he sets: a plot of ground with a gravity and a sun path, one world, or a system of bodies that move one another, a real body taking its values from a cited table and an invented one given by its radius, mass, orbit and spin. A body near enough to matter moves in the physics simulation, a far source of light is a light, and what is too far to move is a backdrop. An environment plan places each top scene's site on that ground at true size and works out the sky each site sees.

A concept for each scene, top down. For each scene a concept is then locked, in order of parent scene and child scene. Each draw is one Nano Banana Pro [29] call that the creator's click pays for, on a concept review page where he marks areas and asks for a new draw. Every draw is sent a **blockout** first, a grey model built in code from the parent's plan at real sizes, so it keeps that layout and proportions and adds the look. A child scene is drawn only once its parent scene is locked and planned, on a crop of the parent's concept, so it fits the parent it stands in.

Inventory and plan. The creator's lock is the approval. Gemini 3.1 Pro lists every object of the locked concept as a row of the scene inventory, with its kind, count, anchor and size in metres, and SAM 3 [23] boxes each row on the concept. A row counts when both find it, or when Gemini names one large object (at least 2 percent of the picture) that SAM 3 misses, boxed by Gemini's region; a row without metres is a gap. Nothing is made or placed that is not a row. The plan is made in two calls. Nano Banana Pro first draws the scene from straight above, from the locked concept, in the frame of the scene's own proportions; Gemini 3.1 Pro then writes the plan in metres from that drawing, the concept and the inventory, element by element: outline, walls, openings, levels, stairs, roofs, terrain heights, each item with its facing, people as the spots they stand on, and the child scenes' rooms. Free checks follow: every row placed as often as it counts, inside the outline, no two floor items in each other, each item on the same side and facing the same way as in the concept, no box round things drawn apart, every door and standing spot reached on foot, and the child scenes tiling their parent. What a check can settle is fixed in code before the plan is shown (items moved apart, turned, added at their place in the concept, split); what remains goes back to Gemini 3.1 Pro as notes at most twice; only what is left is shown as a gap. A plan the creator drew himself wins over a written one, his edits on the plan review page always win and redo what follows, and a child scene he has locked keeps its room when its parent's plan is written again. Only a scene with a design gets the rows of a batch.

Shapes first. The route is one loop, the same at every scale. A parent scene gets its concept, its plan and a rough model, its concept put through picture to 3D whole, which gives the plan its proportions. Its plan is then split into the shapes of its child scenes, and once those shapes are fixed

every child scene is made in parallel, each repeating the loop (the creator's order of 19 October 2026). The runner's rows `rough-model` and `shell-split` (gated on `tiled`, `reached` and `bounded`) are built and have not run.

The scene record. The scene record holds what no made piece covers (shells, stairs, ground and water, backdrop, lights, sky, the neighbours a scene shows), written from the plan, inventory, made ground and sky, and gated on `plan-matched` (the shell within 5 cm of the plan), `openings` (every doorway onto a floor, no view into nothing) and `seed-matched` (every old entry matched or dropped with a reason).

The canonical scene. Every scene is one OpenUSD stage [3]; each object, one per real-world object, carries its kind, transform, collision and each part's library surface. Each department's work is a layer, in a fixed order: a slot for the creator's edit layer on top, then people, sound, light, motion, effects, crops, machines, settle, ground and the generated base. Each scene is assembled once, from every layer's accepted output; its `layers` gate answers unknown while a needed layer has none, and an old output never stands in. A scene shows its neighbours by reference to their own accepted assemblies. The edit layer is still a slot: a test on a two-box scene shows an edit there surviving a regenerated base, but the assembly does not yet carry the creator's edits from one assembly to the next.

Models, generation first. Every object goes the prop route first: Nano Banana Pro draws its close-up once from its crop of the concept, Pixal3D [22] builds it, the surface stage paints it, and a likeness gate compares its outline from eight sides with its close-up. A close-up that fails the shape check is a batch gap, so Pixal3D is never paid for it. Only an object whose checks fail is built by a code builder a coding agent writes (the creator's choice, 12 October 2026). A big object the viewer walks up to gets a body built in code under the generated details its entry names (the creator, 18 October 2026).

Gap: Figures 2 and 3. (2) One scene's design top down: the EC, the parent scene's concept and plan, a child scene drawn on its blockout, its inventory and plan. (3) Objects of one scene from close-up to final model, generated and code-built. They wait on the remake's designs and batch.

3 Stages and their gates

3.1 Surfaces: the finish route

Everything a scene shows comes from the creator's library, score_lib, or is a written exception (the creator's rule of 18 October 2026). Models give shape only: a picture-made model keeps its geometry and its finish is replaced. Every part takes one surface from a shared library of rule-based surfaces, written as ProcFunc functions [30] on Infinigen's shaders [10], coloured from the scene's palette and worn only where wear has a cause (Appendix D).

The finish route gives each model its finishes (the creator's choice of 20 October 2026). Free renders cover every face of the raw model, and each view is split into colour regions. One Gemini 3.1 Pro answer a model lists its finishes, each matched first to a library surface with its settings, and labels each region of each view; the labels vote across views into one finish a face, and a face no label reaches or whose views disagree is a gap. Each part is painted from its library surface and baked, and the bake is checked against the close-up: a finish's surface must lie within ΔE 32 of its colour there, a bake fails when it darkens a finish the close-up shows lighter or turns all surfaces into one grey, and a foot over 3 cm off the floor fails. A free library check traces every material, model, light, sound and character of a scene's accepted assembly to a library entry or a written exception.

3.2 Resting

Loose objects are dropped onto the made ground in a physics simulation; one that would turn more than 15 degrees or drift more than 30 cm is a resting fault of the layout and keeps its laid pose. The settle stage lays where each object comes to rest as a layer (held objects snapped onto their holders, a stack onto one support, an object that would rest worse kept as laid), gated on `rested` and `resting` (no fault that appears with the layer and not without it).

3.3 Light

Lamp strengths are measured. The light layer places live lamps from the scene record, with the sky and its bodies. One paid render per scene gives every lamp its own Cycles light group on the scene's views; because light adds up, the look at any strengths is mixed from these passes without rendering again, and a free sweep chooses the scale that brings the scene's brightness within 1.5 times its concept's, halving any lamp that turns what it lights white.

3.4 Motion, effects and sound

Motion. Each moving thing (a path, a hinge, a spin, a rail, a turn, a climb) has its kind's speed range cited from real machines, gated on `clear` (no moving thing within 3 cm of a still thing or another moving thing, read every quarter second), `speeds`, `loops`, `grounded` and `covered`. **Effects** are emitters measured against real footage, gated on `attached`, `footage`, `light` and `outside` (no open-sky particle inside a room). **Sound:** every source a scene names gets the environment's picked file or one MOSS-SoundEffect [31] take of its brief, gated on `sources`, `level`, `likeness` (CLAP [32] score at least 0.30; a take under it is a gap) and `no freesound`.

3.5 People

People are made like props: a FLUX.2 klein [33] close-up in SOMA-X's bind pose, built by Pixal3D, painted by the surface line, over SOMA-X's body fitted just inside, whose skin weights each look point takes from its closest triangle; Kimodo's clips [34], from sentences, move it. The **shell gate** passes a look when 99 percent of its soft points lie within 5 cm of the body, its worst point recorded; each clip gets its own verdict (tucks, hard parts stretched, cloth or body through cloth), and casts use only passing clips. **Casts:** each person of a scene's inventory is a look of the people line, SAM 3 finds the concept's figures, and casts are gated on `counted`, `standing`, `clear`, `made` and `drapes`.

3.6 Completion

A scene is complete when its gate passes on its final assembly: every row made or dropped with a reason, nothing unmade, no placeholder or resting fault, the creator's picks shown and every layer and shown scene present. A fault the creator decides to ship with is an accepted gap, written with who decided, when, why and its measured faults; the completion gate lists every gap by name and is never loosened, and a scene whose accepted gaps are its only fail is complete apart from accepted gaps. A hook stops an agent claiming a scene is done while its gate fails: agents judging their own scenes agree with a measured geometry score only 45.8 percent of the time [19].

Gap: Figures 4 to 7. (4) A scene's models painted from the library beside their close-ups; (5) a scene swept to its concept's light; (6) motion and effects in place; (7) its people and their cast. They wait on the remake's assemblies.

3.7 Prices and ledgers

Every paid stage is priced from a unit cost measured on the ledgers, times the units its forecast rule counts, times the bill's share over the ledger; a stage with no measured run has no price, and its estimate is shown apart as a guess. Machines and Google calls write ledger lines with their node's clearance, and the month's spend is the higher of ledgers and bill, which caught misses: EUR 1,554 billed against EUR 1,387 in the cloud ledger on 12 October 2026, and EUR 549 billed by Google for a day the earlier ledger put at EUR 30. A paid node runs only when every gate on its inputs passed, the month's books are known and its batch's stop has room for its forecast.

4 Evidence

Environment 1 is the worked example. Its first run was not accepted by the creator and its store was lost on 19 October 2026, so this draft shows none of its outputs; its remake from new designs has not run.

Gap: Table 1 and the film. Per scene of the remake: inventory rows, models generated and coded, layers, gaps before and after the batch, time, euros and the creator's verdict; and a walk through environment 1. They wait on the remake's batch and the creator's review.

Gap: Table 2, people. Usable clips per look by the shell and clip gates on the remake's people. It waits on the people line's batch.

Agent tools. With them an agent used about half the output and cost on two of three scenes; on the third it left more faults; one run each (Appendix E).

5 Limitations, future work and conclusion

One creator judges, and he has not yet accepted a run. Closed models (Nano Banana Pro, Gemini 3.1 Pro) draw the concepts and close-ups and read and plan every scene; they may change or go. The design step, the finish route, the shapes-first loop and the code bodies have not yet carried a scene through a batch; the edit layer is a slot, objects are not interactive, and no game engine opens the result yet. Next: the remake's batch, an engine adapter with physics and interaction, an edit layer the assembly carries, three more environments, open models, LEGO-Bench [19]. In short, SCORE builds an environment the way studios do, scene by scene and layer by layer, so a fault is fixed in its own layer; an accepted environment will show if it holds.

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